

Basic Principles of Medicine 1
Module: Musculoskeletal System | Lecture 28

Histology of the Skin Appendages

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Recommended Reading

Histology – A Text and Atlas

Pawlina 9th Edition

Chapter 15: Integumentary System Pg. 538-587

[https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences-
lwwhealthlibrary-com.sgu.idm.oclc.org/book.aspx?bookid=3290](https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences-
lwwhealthlibrary-com.sgu.idm.oclc.org/book.aspx?bookid=3290)

Images from:

- Pawlina 9th Edition
- Wheater's Functional Histology 5th Edition
- Wheater's Basic Pathology 5th Edition
- Virtual microscopy database by the American Association of Anatomists
- Junqueira's basic histology Text and atlas – <https://accessmedicine-mhmedical-com.periodicals.sgu.edu/book.aspx?bookid=1687>
- <http://www.histology.be/atlas/HSB/index.html>

Overview

The Integumentary System

- Layers of the skin
 - **Dermis**
 - Papillary layer
 - Reticular layer
 - Epidermal ridges & dermal papillae
- Associated Appendages
 - **Hair & hair follicle**
 - Growth cycle
 - **Nails**
 - **Glands**
 - Sebaceous glands
 - Sweat glands
 - Eccrine glands
 - Apocrine glands
- Sensory receptors
 - **Free nerve endings**
 - **Pacinian corpuscles**
 - **Meissner's corpuscle**
 - **Ruffini's corpuscle**
 - **Merkel's corpuscle**

Objectives

Describe the layers of the dermis.	SOM.MK.I.BPM1.2.MSK.3.HCB.1064
Identify the epidermal skin appendages (hair, nails, sebaceous glands, sweat glands).	SOM.MK.I.BPM1.2.MSK.3.HCB.1065
Give the function(s) of hair follicles.	SOM.MK.I.BPM1.2.MSK.3.HCB.1066
Describe the histological structure of hair follicles.	SOM.MK.I.BPM1.2.MSK.3.HCB.1067
Describe the stages of hair growth	SOM.MK.I.BPM1.2.MSK.3.HCB.1068
Describe the microanatomical structure of finger nails.	SOM.MK.I.BPM1.2.MSK.3.HCB.1069
Describe the microanatomical structure of toe nails.	SOM.MK.I.BPM1.2.MSK.3.HCB.1070
Describe the parts responsible for nail growth.	SOM.MK.I.BPM1.2.MSK.3.HCB.1071
Give the function(s) of sebaceous glands.	SOM.MK.I.BPM1.2.MSK.3.HCB.1072
State the localization of sebaceous glands.	SOM.MK.I.BPM1.2.MSK.3.HCB.1073
Describe the microscopic structure of sebaceous glands.	SOM.MK.I.BPM1.2.MSK.3.HCB.1074
Give the function(s) of eccrine sweat glands.	SOM.MK.I.BPM1.2.MSK.3.HCB.1075
State the localization of eccrine sweat glands.	SOM.MK.I.BPM1.2.MSK.3.HCB.1076
Describe the microscopic structure of eccrine sweat glands.	SOM.MK.I.BPM1.2.MSK.3.HCB.1077
Give the function(s) of apocrine sweat glands.	SOM.MK.I.BPM1.2.MSK.3.HCB.1078
State the localization of apocrine sweat glands.	SOM.MK.I.BPM1.2.MSK.3.HCB.1079
Describe the microscopic structure of apocrine sweat glands.	SOM.MK.I.BPM1.2.MSK.3.HCB.1080

Objectives

Compare and contrast the mode of secretion in sebaceous glands vs. eccrine sweat glands vs. apocrine sweat glands.

SOM.MK.I.BPM1.2.MSK.3.HCB.1081

List the function(s) of free nerve endings in the epidermis.

SOM.MK.I.BPM1.2.MSK.3.HCB.1082

Describe the microscopic structure of free nerve endings in the epidermis.

SOM.MK.I.BPM1.2.MSK.3.HCB.1083

List the function(s) of Pacinian corpuscles.

SOM.MK.I.BPM1.2.MSK.3.HCB.1084

State the localization of Pacinian corpuscles.

SOM.MK.I.BPM1.2.MSK.3.HCB.1085

Describe the microscopic structure of Pacinian corpuscles.

SOM.MK.I.BPM1.2.MSK.3.HCB.1086

List the function(s) of Meissner's corpuscles.

SOM.MK.I.BPM1.2.MSK.3.HCB.1087

State the localization of Meissner's corpuscles.

SOM.MK.I.BPM1.2.MSK.3.HCB.1088

Describe the microscopic structure of Meissner's corpuscles.

SOM.MK.I.BPM1.2.MSK.3.HCB.1089

List the function(s) of Ruffini's corpuscles.

SOM.MK.I.BPM1.2.MSK.3.HCB.1090

State the localization of Ruffini's corpuscles.

SOM.MK.I.BPM1.2.MSK.3.HCB.1091

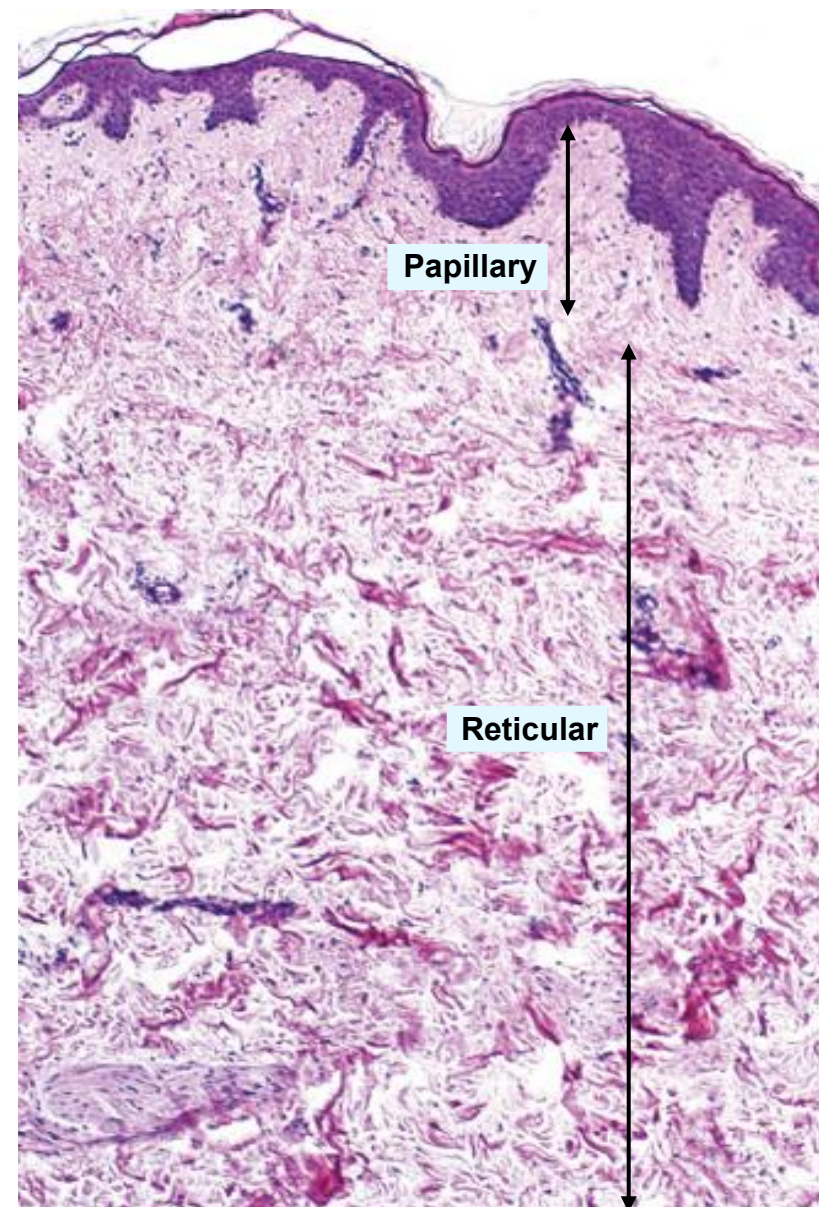
Describe the microscopic structure of Ruffini's corpuscles.

SOM.MK.I.BPM1.2.MSK.3.HCB.1092

The Dermis

Underlying connective tissue that provides support, strength and elasticity

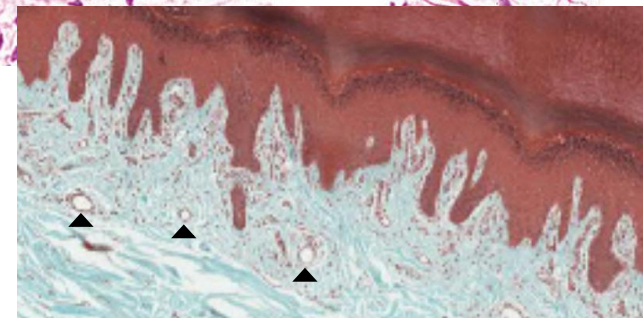
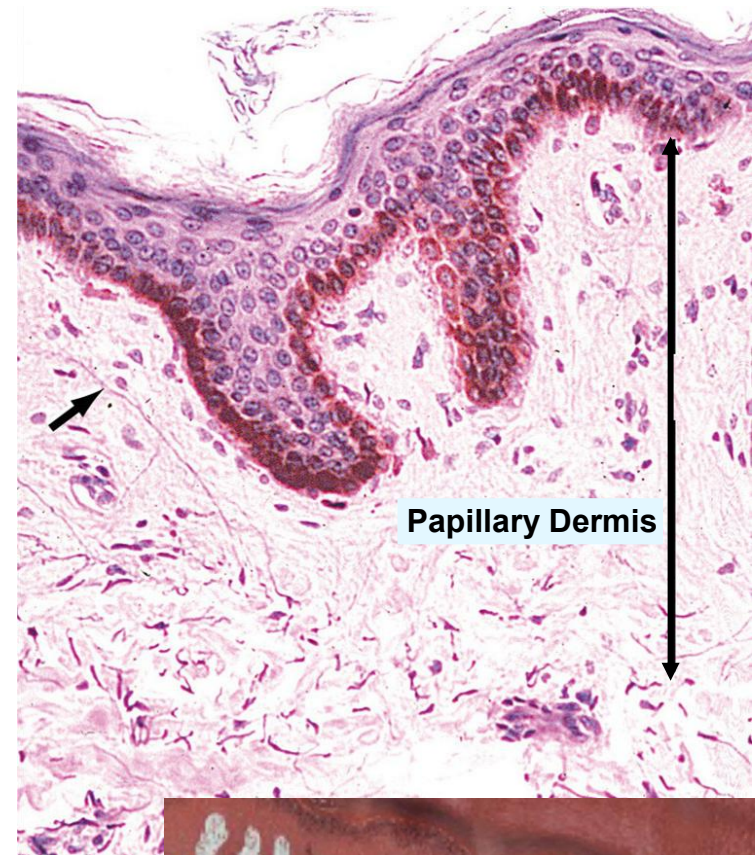
- Contain many epidermal appendages and sensory receptors
- Composed of 2 major layers:
 - **Papillary layer** – Loose connective tissue
 - **Reticular layer** – Dense connective tissue
 - Around the areolae, penis, scrotum and perineum, smooth muscle cells form a loose plexus in the reticular layer
 - This accounts for the **puckering of the skin** at these sites, particularly in erectile organs



Papillary Dermis

Superficial layer of areolar connective tissue (cells > fibers)

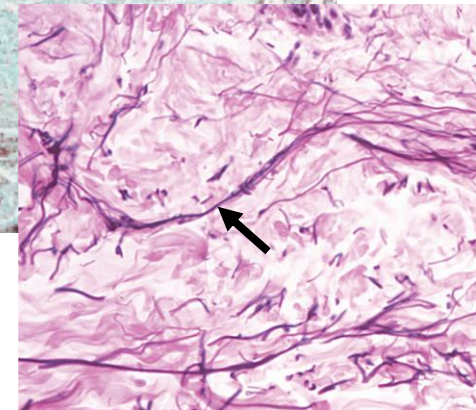
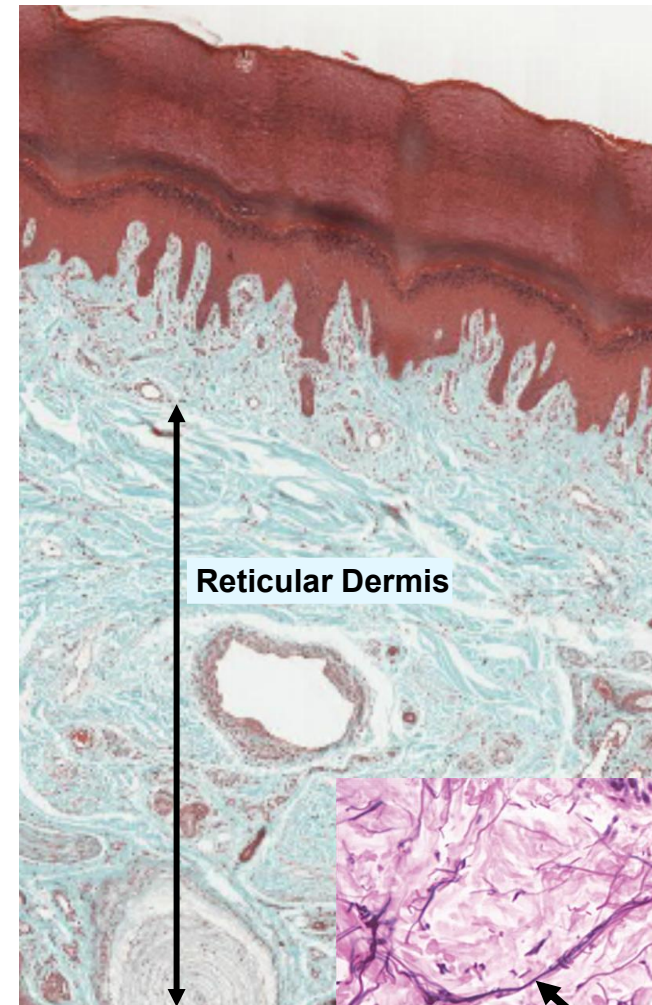
- **Highly vascularized**
- Predominately type I and III collagen fibers
- The elastic fibers are threadlike and form an irregular network
- Relatively thin and includes the substance of the **dermal papillae** and **dermal ridges**
- Contains **Meissner's corpuscles**



Reticular Dermis

Deep layer of dense irregular connective tissue (fibers > cells)

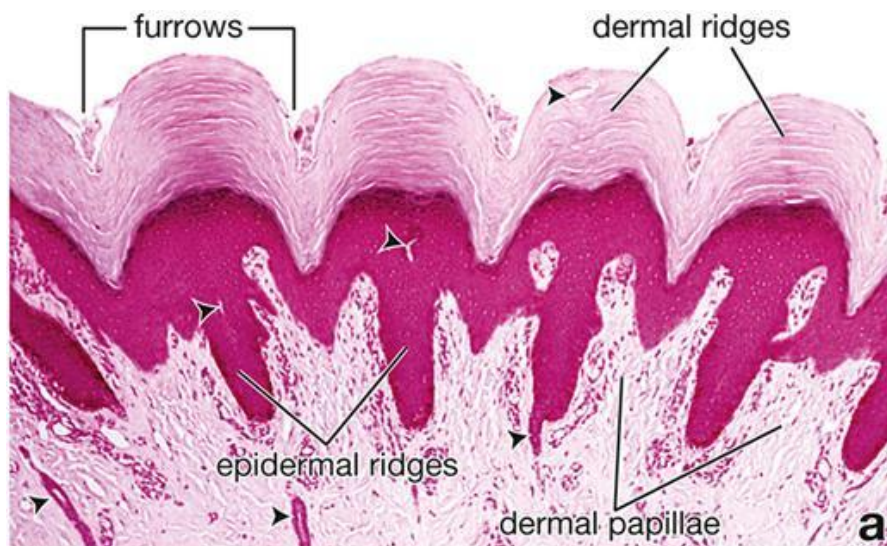
- **Vascularized**
- Considerably thicker than the papillary layer
- Characterized by thick, irregular bundles of predominantly type I collagen fibers and coarser elastic fibers
 - These fibers are not randomly oriented but instead form **regular lines of tension**



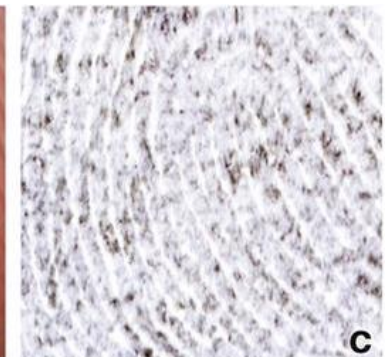
Epidermal ridges & Dermal papillae/ridges

At the **dermal-epidermal junction** is an uneven boundary of numerous finger-like protrusions of the dermis and epidermis

- **Epidermal ridges** → epidermal protrusions that project into the dermis
- **Dermal papillae** → connective tissue projections into the undersurface of the epidermis
- **Dermal ridges** → parallel arrangement with the dermal papillae on palms and soles
 - Responsible for unique **fingerprinting**
 - Visible ridges on the surface of the skin



Areas of increased mechanical stress has deeper epidermal ridges and longer, more closely spaced dermal papillae

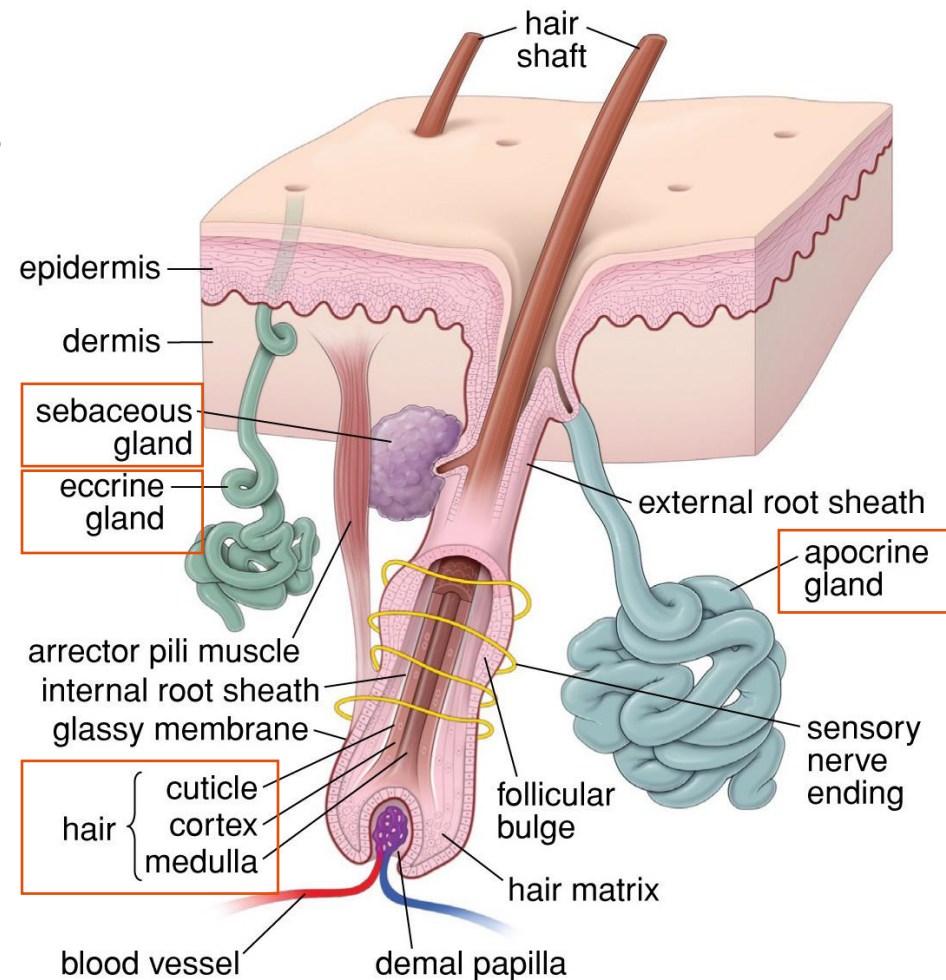


Associated Appendages

Down growths of the epidermis into the dermis during embryological development

- Modified to perform specific duties

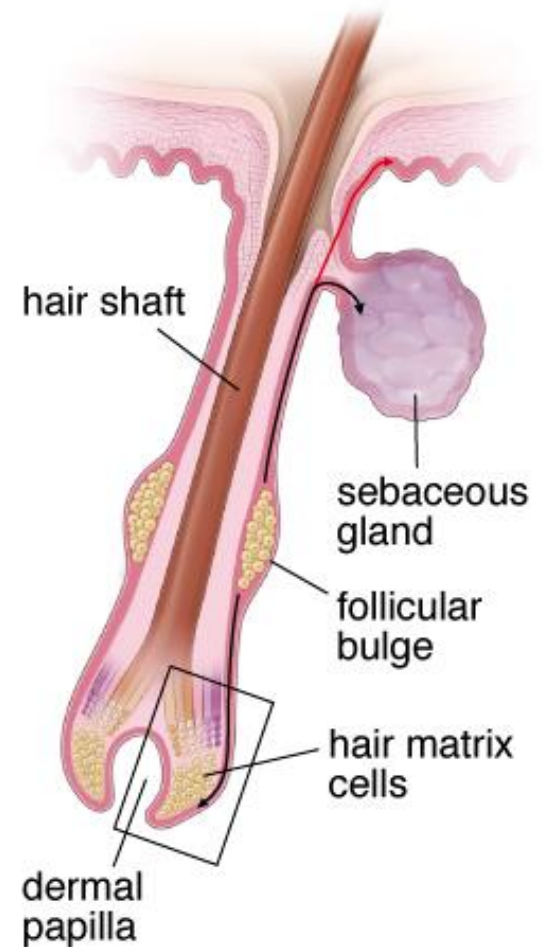
- Hair & hair follicle**
- Nails**
- Glands**
 - Sweat
 - Sebaceous
 - Mammary



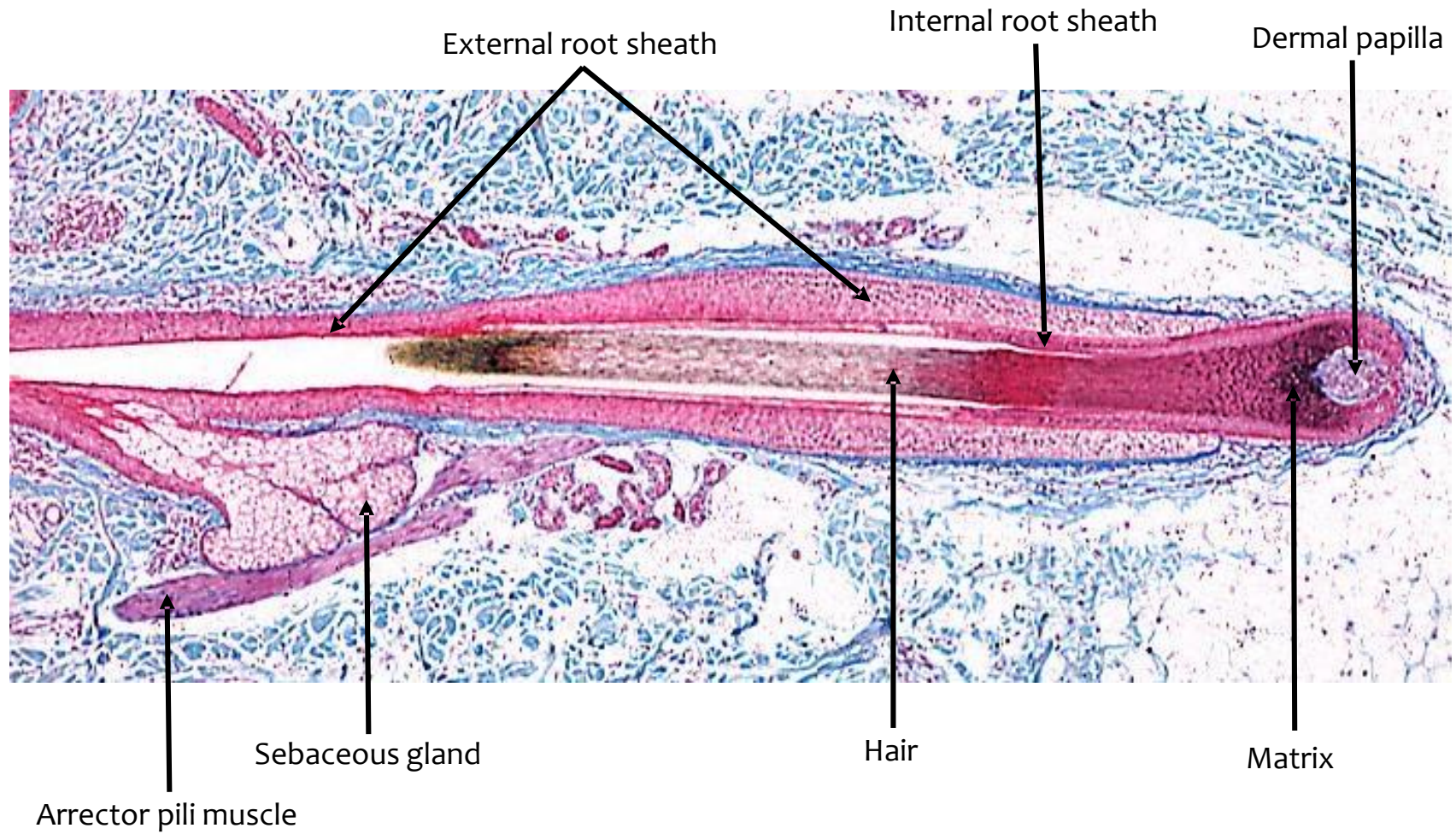
Hair

Cylindrical shafts composed of keratin

- Produced in follicles that extend into the hypodermis
- Found in **thin skin only**
- Associated with sebaceous glands
 - **Pilosebaceous organ/unit** → hair follicle plus sebaceous gland and arrector pili muscle
- Consists of 4 major parts:
 - **Infundibulum**
 - **Isthmus**
 - **Follicular bulge**
 - **Inferior segment**
 - Includes the **Bulb** which contains **hair matrix cells** and **melanocyte stem cells**



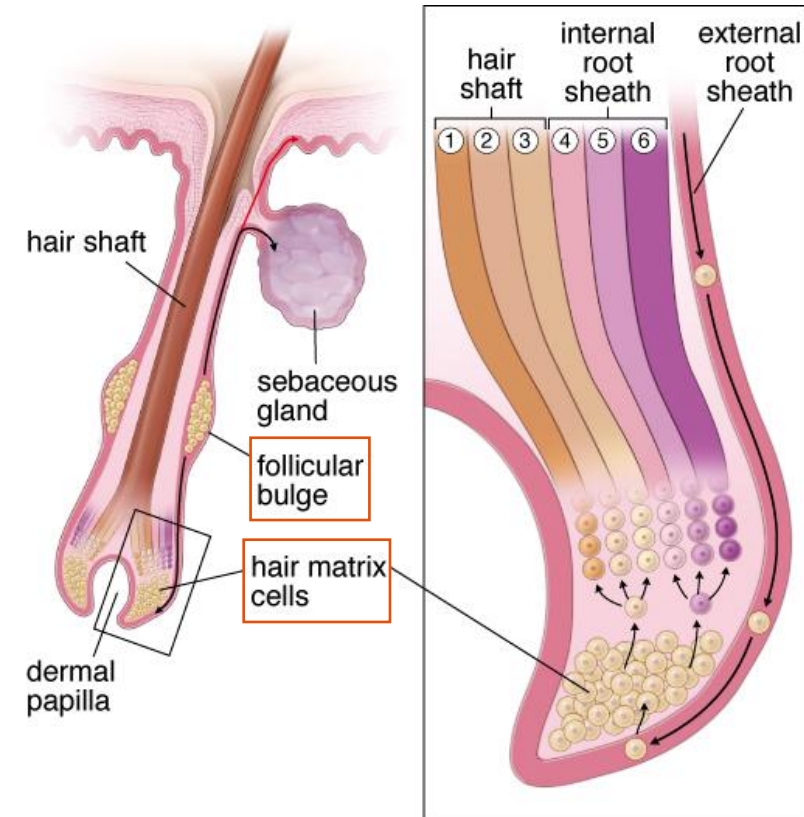
Hair



Hair Follicle & Hair Formation

Within the **follicular bulge** is an aggregate of epidermal stem cells

- These cells are responsible for the growth of hair follicle and hair
- Keratinization of the internal root sheath and hair shaft occurs within the **keratogenous zone**
 - As the cells pass through this zone, they differentiate, lose their organelles, and become tightly packed with **hard keratin**
 - The fully keratinized hair is not followed by the internal root sheath as it grows
 - A thick basal lamina, called the **glassy membrane**, separates the hair follicle from the dermis



Hair



Hair Growth

- Cyclical process that occurs in 3 phases:
 - **Anagen** – Long phase of active growth
 - **Catagen** – Short phase of involution
 - **Telogen** – Short inactive phase
- Growth cycle varies from site to site:
 - Scalp hair follicles have a long anagen growth phase of 2 years and a short telogen phase of a few months
 - Pubic hair, trunk hair, eyelashes and eyebrows have a short growth phase and a relatively long resting phase hence they are not as long

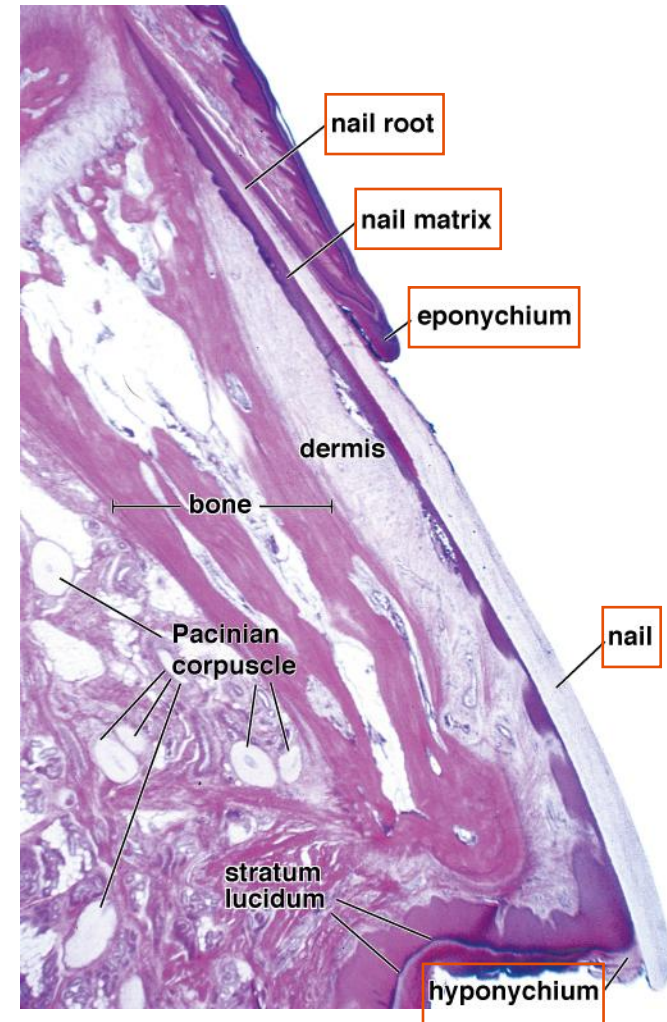
Nails

Hard plates of keratinized cells present on the dorsal surface of the distal phalanx of the fingers and toes

- Consists of 2 major parts:
 - Nail plate** – the visible portion secured distally by a thickened epidermal layer known as the **hyponychium**
 - Rests on the **nail bed** which consists of epithelial cells continuous with the stratum basale and stratum spinosum
 - Nail root** – the proximal portion of the nail plate covered by the **eponychium**
 - The **nail matrix** is located deep to the nail root and contains stem cells which regularly divide and migrate toward the root of the nail facilitating growth

Functions:

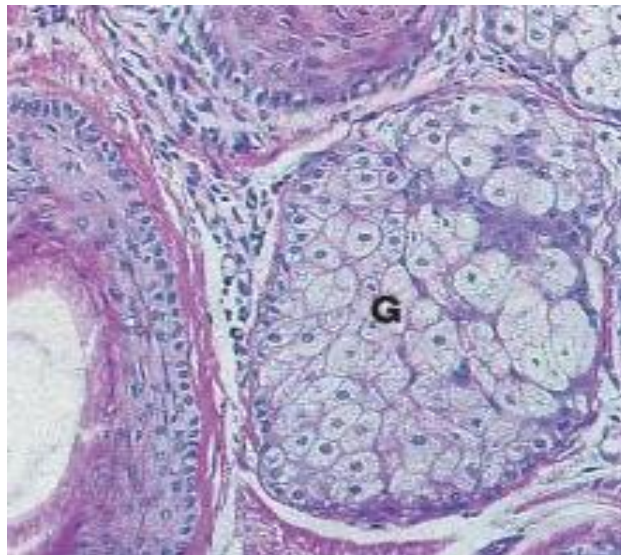
- Protection
- Helps with grip



Sebaceous Glands

Simple branched acinar gland found in thin skin only

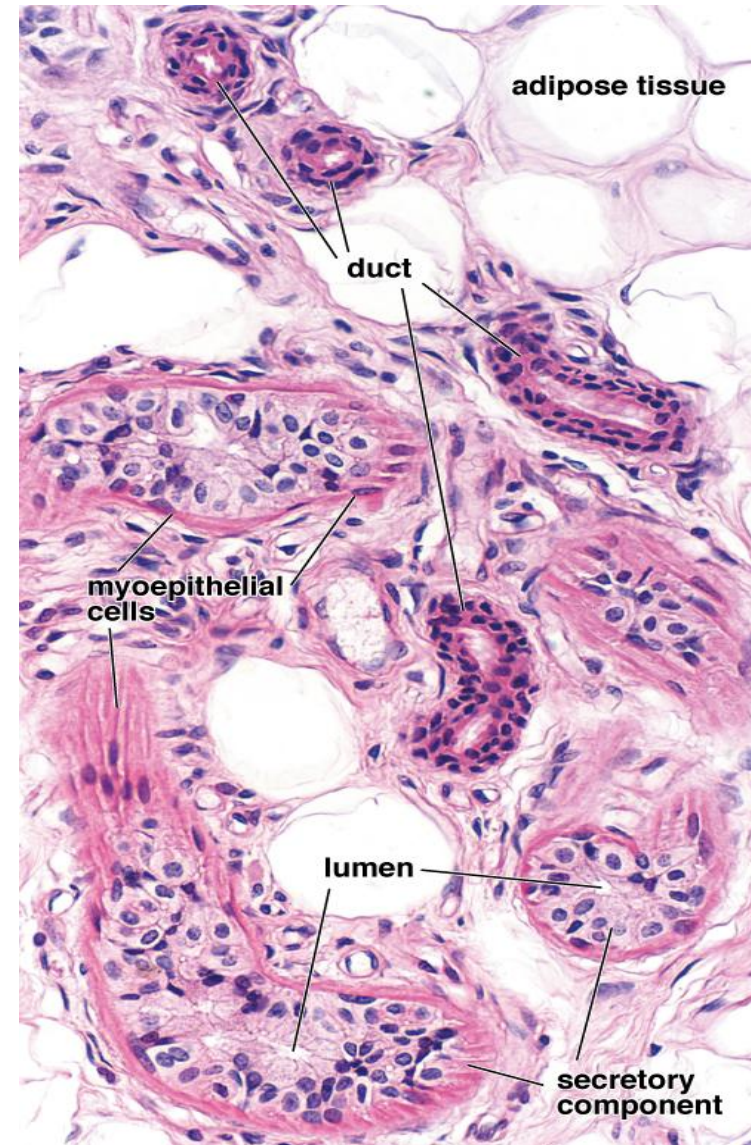
- Secretory portion is located deep in the dermis
 - Produces an oily/fatty substance known as **sebum**, released via **holocrine secretion**
 - Stains poorly with H&E
- Ductal portion opens into the **hair follicle**
 - Prevents the hair from becoming dry and brittle
 - Clinical correlate: **Acne**



Eccrine Glands

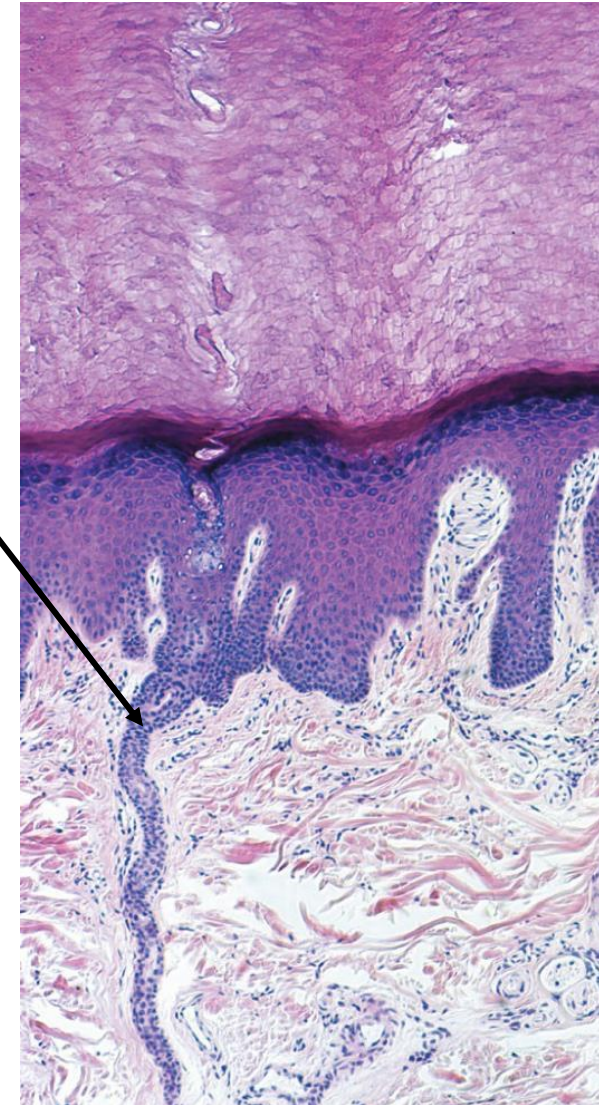
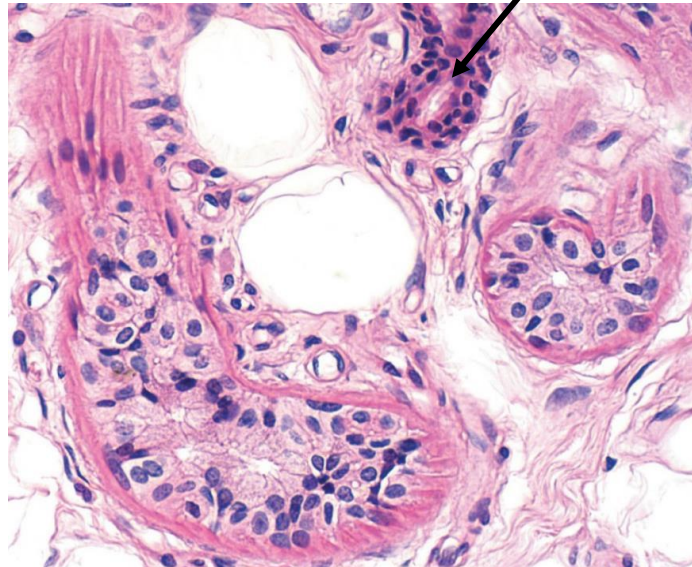
Simple coiled tubular gland found in thick **and** thin skin

- Innervation – **cholinergic**
- Secretory portion is located deep in the dermis
 - Produces a watery secretion – **sweat**, released via **merocrine secretion**
 - Lined by **pseudostratified cuboidal epithelium** containing 3 major cell types:
 - **Clear cells** – abundant glycogen
 - Stains well with PAS
 - **Dark cells** – abundant rER and secretory granules
 - **Myoepithelial cells** – located at the basal aspect of the secretory segment



Eccrine Glands

- Ductal portion opens onto the **surface of the skin**
 - Gentle spiral course to the epidermis which becomes tighter toward the surface
 - Lined by **stratified cuboidal epithelium** consisting of a basal layer and a luminal layer



duct

Apocrine Glands

Simple coiled tubular gland found in thin skin only

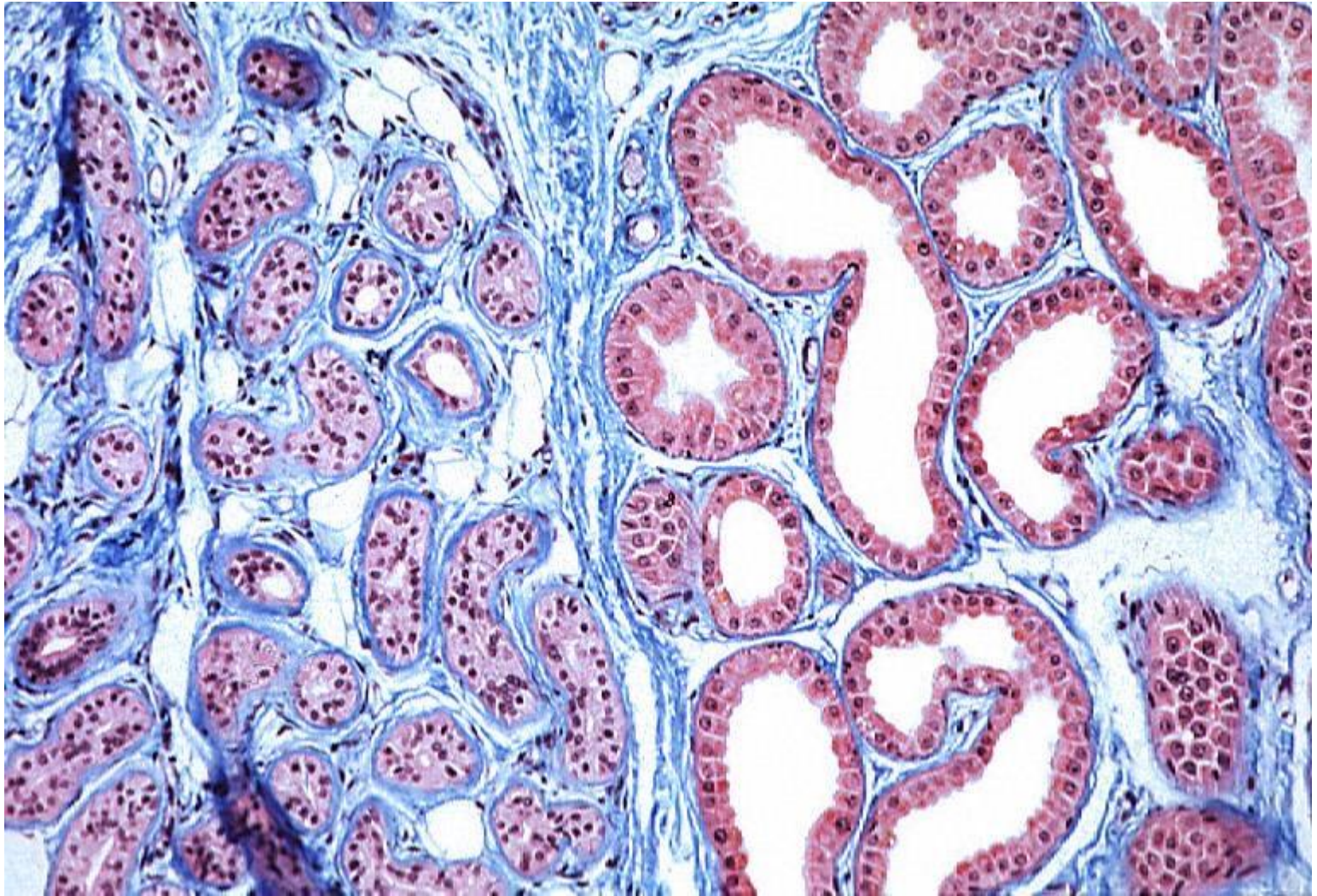
- Located in the **pubic regions**
- Innervation – **adrenergic**
- Secretory portion is located deep in the dermis and hypodermis
 - Produces a viscous secretion containing **pheromones**, released via **merocrine secretion**
 - Lined by **simple cuboidal to columnar epithelium**
 - Lumen of the secretory portion are larger than eccrine sweat glands
- Ductal portion opens into the **hair follicle**
 - Straight with a narrow lumen
 - Lined by **stratified cuboidal epithelium**



Eccrine Glands vs. Apocrine Glands

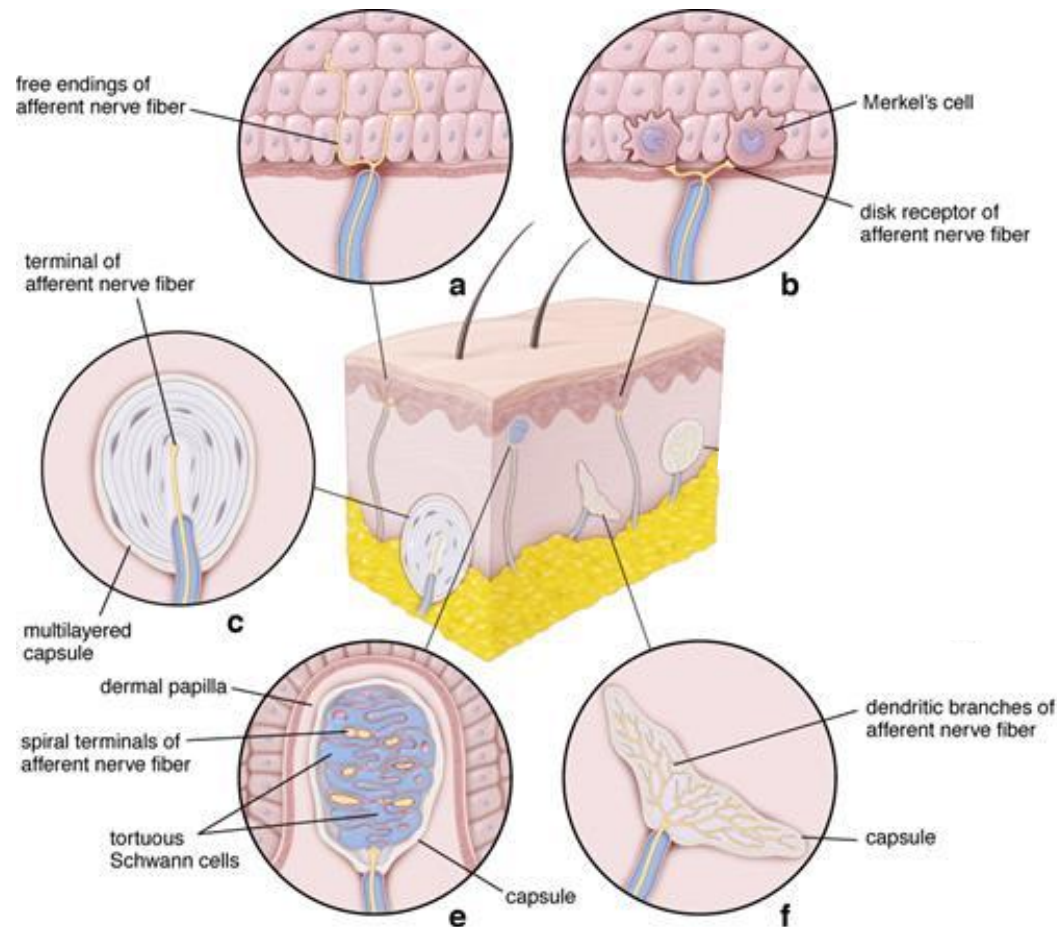
Features	Eccrine	Apocrine
Location	Thick & thin skin	Thin skin only • Axilla • Areolas • Perineum • Perianal region
Luminal size	Small	Large
Secretory product	Watery	Viscous
Secretion mode	Merocrine	Merocrine
Duct	Coiled and opens onto the surface	Straight and opens into hair follicles
Innervation	Cholinergic	Adrenergic
Function	Thermoregulation	Pheromones

Eccrine Glands vs. Apocrine Glands



Nerve Supply

- **Sensory receptors** – peripheral terminals of sensory nerves
- **Secretomotor nerve endings** to the blood vessels, arrector pili muscles and sweat glands



Free Nerve Endings

Occupy the papillary dermis and send branches into the epidermis that terminate in the stratum granulosum

- Most numerous
- Lack myelin or connective tissue coverings
- Functions
 - Subserve **multiple sensory modalities**
 - **Pain, temperature, itching, fine touch**
 - Networks of free dermal endings surround most hair follicles
 - Attach to their outer root sheath
 - Very sensitive to hair movement and serve as **mechanoreceptors**
 - This relationship imparts a sophisticated degree of specialization in the receptors that surround tactile hairs as is found in the whiskers of certain animals in which each represents its own area on the cortex

Pacinian Corpuscle

Large, ovoid structures located deep in the dermis and hypodermis

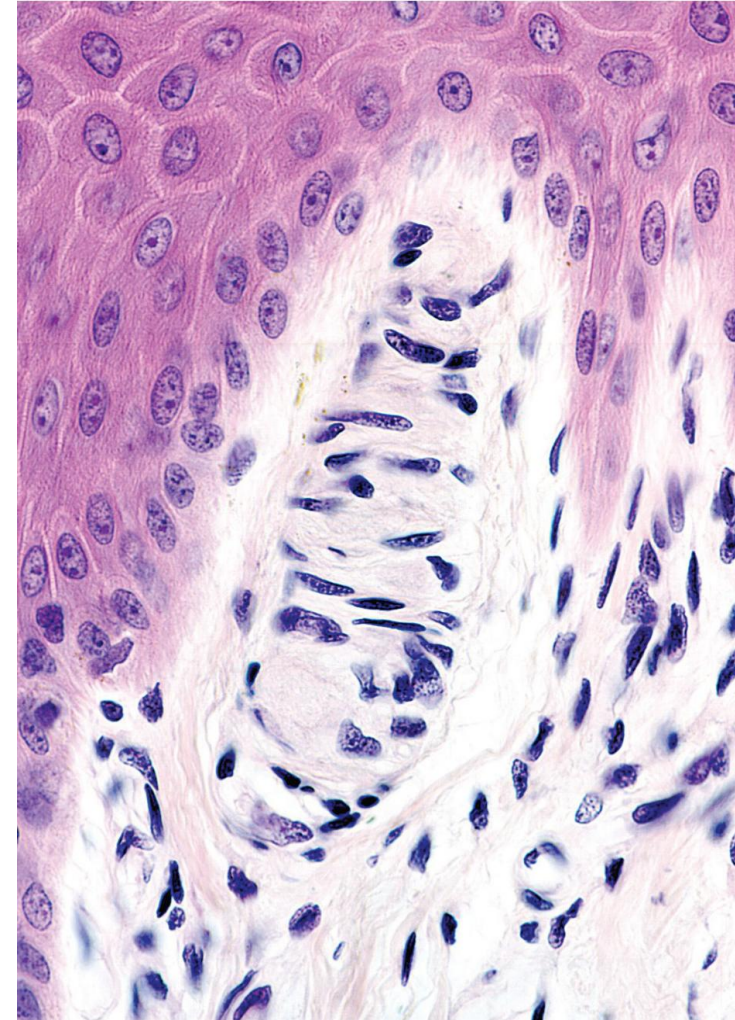
- Have macroscopic dimensions more than 1mm along their long axis
- Composed of a myelinated nerve ending surrounded by a capsule structure
- Nerve loses its myelin sheath after entering capsule
 - The unmyelinated portion is covered by a series of tightly packed, flattened Schwann cell lamellae that form the inner core of the corpuscle
 - The remainder is formed by a series of concentric lamellae
- Functions
 - **Pressure** and **vibration**



Meissner's Corpuscle

Tapered cylinders located in the dermal papillae just beneath the epidermal basal lamina

- Appears as twisted **skein of wool**
- Perpendicular to the skin surface
- 150 microns along their long axis
- There are one or two unmyelinated nerve endings of myelinated nerve fibers that follow spiral paths in the corpuscle
- The cellular component consist of flattened Schwann cells that form several irregular lamellae through which the axons course to the pole of the corpuscle
- Functions
 - **Touch receptors** particularly responsive to **low frequency stimuli** (light touch)



Ruffini's Corpuscle

Elongated fusiform shape

- 1-2 microns along their long axis
- Composed of a single myelinated fiber with a thin connective tissue capsule surrounding a fluid filled space
 - Fibers lose its myelin sheath as it enters the capsule and branches to form a dense arborization of fine axonal endings each terminating in a small knoblike bulb
 - The axonal endings are dispersed and intertwined inside the capsule
 - Collagen fibers from surrounding connective tissue pass through the capsule
 - The axonal endings respond to displacement of the collagen fibers induced by sustained or continuous mechanical stress
- Functions
 - **Stretch** and **torque**

Merkel's Corpuscle

Merkel's cell in communication with expanded terminal bulb of afferent myelinated nerve fibers → Merkel's corpuscle

- Cytoplasm contains **neurosecretory granules**
- Express antigenic markers of both epidermal and neural type
- Function
 - Most abundant in skin where sensory perceptions is acute (fingertips)
 - Detect **touch sensations**



Thank you!



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